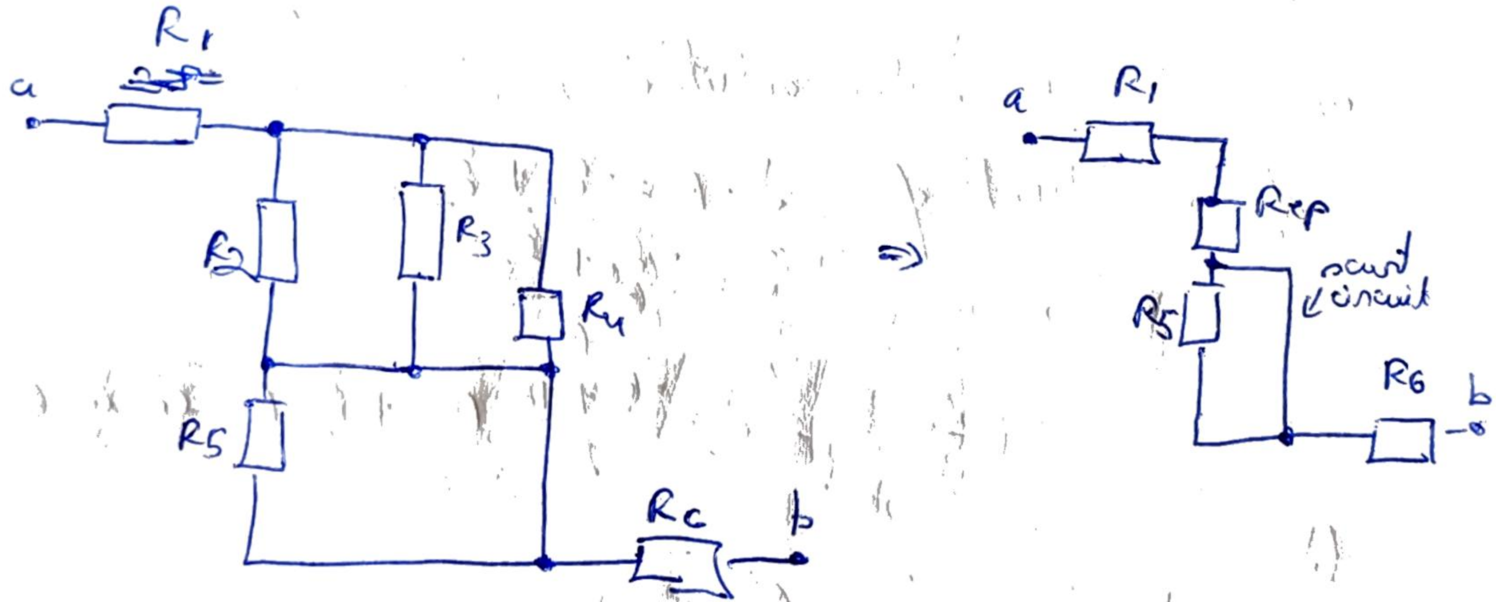
4'/817



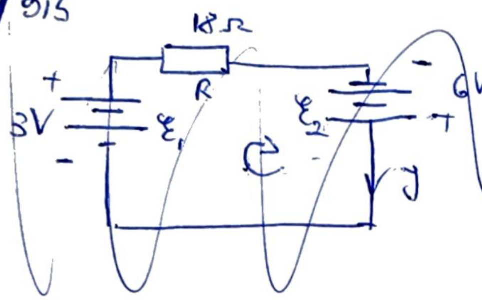
$$R_{ep} = \frac{1}{\frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}} = \frac{1}{\frac{2}{6}\!\!\!\!\!\;\frac{1}{6} + \frac{1}{12} + \frac{3}{4}\!\!\!\!\;\frac{1}{4}} = \frac{1}{\frac{6}{12}} = 2\ \Omega$$

$$R_{ec} = R_1 + R_{ep} + R_6 = (2 + 2 + 3)\ \Omega = 7\ \Omega$$

(R_{ab})

1/

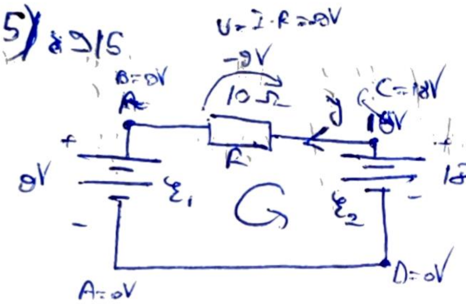
4) 2015



a) J = ?, direcția curentului

b) ~~$P(d) = V U$~~

5) 2015



a) J = ?, direcția curentului

b) ~~$P(d) = V$~~;

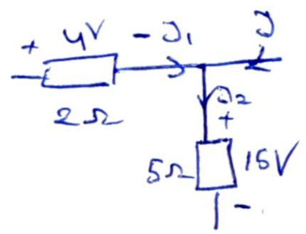
a) Se alege un sens arbitrar pentru curent

Lg II Kirchhoff: $J \cdot R = -\mathcal{E}_1 + \mathcal{E}_2$

$$J = \frac{-\mathcal{E}_1 + \mathcal{E}_2}{R}$$

$$J = \frac{-9+18}{10} A = +0{,}9\ A$$, ccw

3/2015



J = ?, direcție J

Lg:

$$J_1 = \frac{4V}{2\Omega} = 2A$$

$$J_2 = \frac{15V}{5\Omega} = 3A$$

Lg I K: $J_2 = J_1 + J \Rightarrow J = J_2 - J_1 = (3-2)A = 1A$

spre stânga